Siddhant Prakash

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linkedin.com/in/SiddhantPrakash — github.com/SiddhantPrakash — Portfolio

SUMMARY

Final year IT student with a strong focus on logical reasoning, Data Engineering, and AI-driven software development. Proven track record in research-based machine learning applications, data processing pipelines, and database architecture. Eager to contribute to high-scale enterprise platforms and analytics solutions by leveraging expertise in C++, SQL, javascript, and data-driven problem solving.

EDUCATION

National Institute of Technology, Srinagar — *B.Tech in Information Technology (CGPA: 8.8/10)* 2023 – 2027
D.A.V. Public School, Barkakana — *Class XII, CBSE (93.2%)* 2022
St. Ann's School, Ramgarh — *Class X, ICSE (95.6%)* 2020

TECHNICAL SKILLS

Languages (Preferred): Python, C++, JavaScript
Machine Learning & Analytics: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
Databases & Cloud Tech: MongoDB, MySQL, Cloudinary, REST APIs
Web Development: HTML5, CSS3, React.js, Node.js, Express.js
Tools & Fundamentals: Git, GitHub, VS Code, Jupyter Notebook

COURSEWORK

Core Subjects: Machine Learning, Artificial Intelligence, Database Management Systems (DBMS), Operating Systems, Computer Networks

EXPERIENCE

Research Intern — Indian Institute of Technology (IIT) Mandi — Dec 2025 – Feb 2026 Certificate

- Worked on research project *PPG to ECG Signal Conversion using Convolutional Neural Networks*
- Evaluated deep learning architectures for complex biomedical signal reconstruction
- Conducted rigorous quantitative analysis using MAE, RMSE, R^2, Pearson Correlation, DTW, and SNR metrics to drive data-backed model selection

PROJECTS

Customer Segmentation & Decision Support System — Python, Scikit-learn — GitHub

- Applied K-Means clustering algorithms to segment mall customers based on purchasing behavior and income streams
- Processed, cleaned, and analyzed a dataset of 200+ records using Pandas and NumPy to identify underlying patterns
- Visualized customer clusters using Matplotlib and Seaborn to generate actionable business insights and support targeted strategic decisions

Full-Stack Airbnb Clone — MongoDB, Express.js, React.js, Node.js, Cloudinary — GitHub

- Developed a comprehensive accommodation booking platform featuring robust user authentication, property listing management, and secure booking workflows
- Architected RESTful APIs using Node.js and Express to handle seamless data flow between the interactive React frontend and MongoDB database
- Integrated Cloudinary for seamless cloud-based image uploading and management, optimizing media storage and delivery for property listings
- Implemented dynamic UI components and centralized state management to deliver a highly responsive and intuitive user experience across devices

GenAI Content Generator — Python, LangChain, Streamlit, REST APIs — GitHub

- Developed a scalable web application using Streamlit and LangChain to generate customized professional content based on user-defined parameters.
- Integrated Groq's Llama-3 API for rapid text generation, optimizing API calls and utilizing few-shot prompting to accurately replicate specific output styles.
- Engineered a resilient data processing pipeline using Pandas to parse JSON datasets, implementing custom algorithms to handle edge cases and bypass broken unicode characters.

CAMPUS INVOLVEMENT

Web Developer — Range-Chinaar Technical Team — 2023

- Designed responsive layouts for an event portal serving 500+ students, improving digital engagement metrics

Technical Team Member — Techvaganza Technical Fest — 2024

- Coordinated technical logistics and problem-solved real-time challenges for a large-scale college fest with 1000+ participants